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# Morphological volatility precedes ecological innovation in early echinoderms

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**Origins of higher taxonomic groups entail dramatic and nearly simultaneous changes in morphology and ecological function, limiting our ability to disentangle the drivers of evolutionary diversification. Here we phylogenetically compare the anatomy and life habits of Cambrian–Ordovician echinoderms to test which facet better facilitates future success. Rates of morphological evolution are faster and involve more volatile trait changes, allowing morphological disparity to accrue faster and earlier in the Cambrian. However, persistent life-habit evolution throughout the early Palaeozoic, combined with iterative functional convergence within adaptive strategies, results in major expansion of ecospace and functional diversity. The interactions between tempo, divergence and convergence demonstrate not only that anatomical novelty precedes ecological success, but also that ecological innovation is constrained, even during a phylum's origin.**

The origins of phyla and other higher taxonomic groups are fundamentally ecological events fostered by lax competition with concomitant opportunities for developmental novelty[1–5], and associated with elevated rates of both morphological evolution and ecological change[6–13]. However, because most studies utilize the same underlying data to interpret both evolutionary patterns, it is difficult to determine whether ecological or morphological novelty is the primary driver of a clade's success[2,13,14]. Using independently compiled datasets, here we compare the anatomy and life habits of all known early Palaeozoic echinoderm lineages to illuminate how taxonomic diversifications are manifested morphologically and ecologically. More generally, does morphological novelty drive later ecological success, or vice versa?

Echinoderms underwent their greatest morphologic, ecologic and taxonomic diversification during the Cambrian and Ordovician radiations[15,16]. More than 20 taxonomic classes (including all extant ones) and some 50 orders originated by the end of the Ordovician (Extended Data Fig. 1)[17–20]. The analyses presented here are based on a stratigraphically calibrated phylogeny of 366 early Palaeozoic echinoderm tip genera and independently compiled datasets of 413 anatomical[16] and 40 life-habit traits[21] that comprehensively characterize the form and function of these echinoderms, including inferred states for phylogenetic common ancestors. The inclusion of independent datasets allows for an explicit test of whether morphological traits provide a different evolutionary signal than do ecological (functional) traits[13].

## Results

**Morphological and ecological richness trends.** Lineage richness (tips and inferred common ancestors; Extended Data Fig. 1) and morphological richness (number of unique morphotypes; Fig. 1a) increase rapidly by the mid Cambrian, increase more rapidly into the Early Ordovician, and persist at high but slightly reduced levels throughout the period. (The Late Ordovician decline is primarily an artefact because our phylogenetic data do not extend the ranges of post-Ordovician taxa and their ancestral lineages[22]; Extended Data Fig. 1.) The Cambrian diversifications produced distinct morphological clusters (Fig. 2), roughly corresponding to four recognized subphyla[16,18,23]. These earliest echinoderms were already ecologically diverse (Fig. 2), exhibiting a range of feeding habits, substrate preferences, microhabitat specializations, motilities, body sizes and other functionally important life habits[24,25]. When standardized to a common genus richness (Extended Data Figs. 2 and 3), life-habit richness (number of unique life habits) increases every interval. The raw (non-standardized) trend in life-habit richness is similar to the trend in number of morphotypes (Fig. 1a; detrended correlation coefficient $r = 0.95$, $P = 9.27 \times 10^{-9}$), but the rate of diversification is dampened, especially in the Cambrian. Peak Cambrian morphological richness occurs in Stage 4 before declining, whereas ecological richness plateaus during the mid Cambrian and peaks 24 Myr later, before dramatic increases in both trends during the Ordovician, when both peak during the same interval.

**Morphology evolves at a faster rate than does ecology.** Documented using multiple discrete-character tests[26,27], the rate of character change in both datasets is relatively constant throughout the early Palaeozoic (Fig. 1b), with pulses that coincide with the major intervals of lineage diversification. This coincidence in timing suggests that the Cambrian and Ordovician radiations were associated with both anatomical and ecological novelty involving multiple echinoderm classes (Extended Data Fig. 1). The tempo of morphological evolution is, on average, 71% greater than the rate of ecological change, and whether measured across the phylogeny through time (Fig. 1b) or on a per-character basis (Fig. 3a). The difference is especially pronounced during the mid Cambrian, allowing morphological diversity to quickly outpace functional diversity (Fig. 1). The average rate continues to increase slightly throughout the Middle-to-Late Ordovician, suggesting that the evolution of novel anatomies and life habits continues unabated throughout the early Palaeozoic.

The elevated morphological rate is not an artefact of fundamental differences in the structure of each dataset. Although the morphological dataset contains greater hierarchical character-dependence

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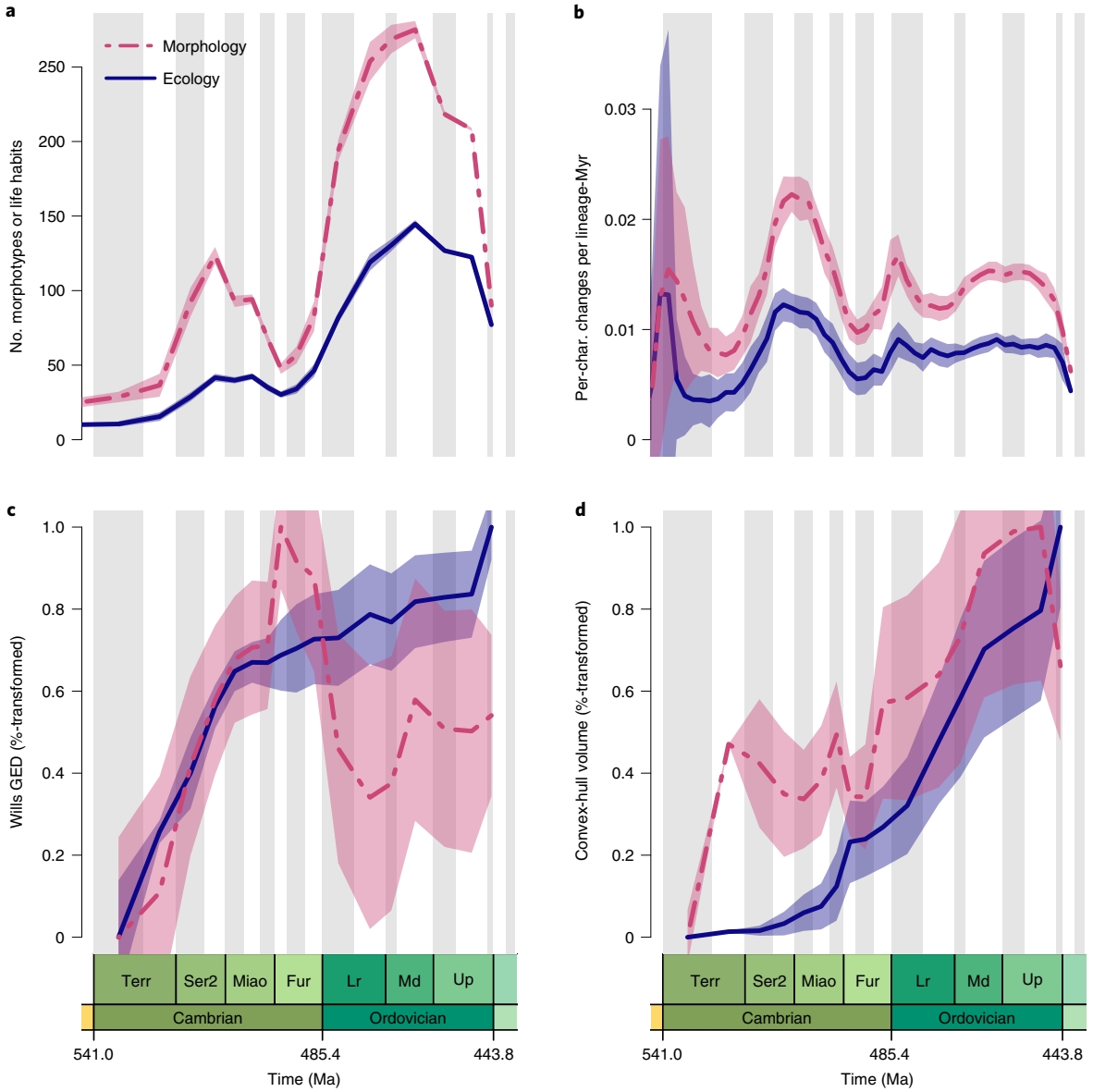



**Fig. 1 | Distinct trends in echinoderm morphology and ecology during the early Palaeozoic. a**, Number of unique morphotypes and life habits ('mode' treatment), including tips and inferred ancestors. See Extended Data Fig. 1 for lineage richness trends. **b**, Rates of discrete per-character change. Likelihood tests confirm that the datasets evolve at different rates ($\lambda_{morphology} = 6.15$ character changes per lineage-Myr; $\lambda_{ecology\text{-}mode} = 3.59$, $\lambda_{ecology\text{-}constant} = 2.93$, $\lambda_{ecology\text{-}raw} = 3.08$; Akaike weight$_{2\text{-}rate\ model} = 1.000$ for all 50 time-scaled phylogenies, for all data treatments). **c**, Sample-standardized (50 lineages per interval) pairwise morphological and ecological disparity calculated using Wills GED, with alpha = 0.5 to balance the influence of character dependence; trends are not correlated (detrended correlation coefficient $r = 0.318$, $P = 0.247$). **d**, Convex-hull hypervolume calculated across the first six principal coordinate axes; trends are significantly negatively correlated (detrended correlation coefficient $r = -0.546$, $P = 0.035$). For **a** and **b**, trends and standard errors were estimated as the mean and standard deviation across 50 time-scaled phylogenies; for **c** and **d**, trends and standard errors were calculated from 3,000 resampling replicates (calculating statistics on random samples of 50 lineages per bin for each of 50 time-scaled phylogenies, replicated 60 times), and percent-range transformed to focus on trend comparisons. The raw (non-sample-standardized) values for the Fortunian (first stage of Terreneuvian) are included for **c** and **d** to allow an estimate of their earliest disparity (see Extended Data Fig. 3 for trends without this age included). The mean and standard error for this age are estimated from the mean and standard deviation across 50 time-scaled phylogenies.

structure—that is, a change in one state can compel subsequent changes in subordinate characters[28]—there is insignificant variation among the rates at which hierarchically dependent characters evolve compared with their linked characters, in either dataset (Extended Data Fig. 5). Considered in isolation, primary characters in the morphological dataset actually evolve faster than their dependents[16], a pattern that is reversed in the ecological dataset. The elevated pace of primary morphological character evolution is consistent with continuing changes to the fundamental anatomical architecture of these early echinoderms[16]. The rate differences are compounded because many ecological character states have equally complicated character covariances with other states as do morphological characters, especially those related to body size[21]. Per-character rate calculations are sensitive to the number of characters and character states[27,29], and potentially taxon-sampling heterogeneities through time, but subsampling analyses rule out these

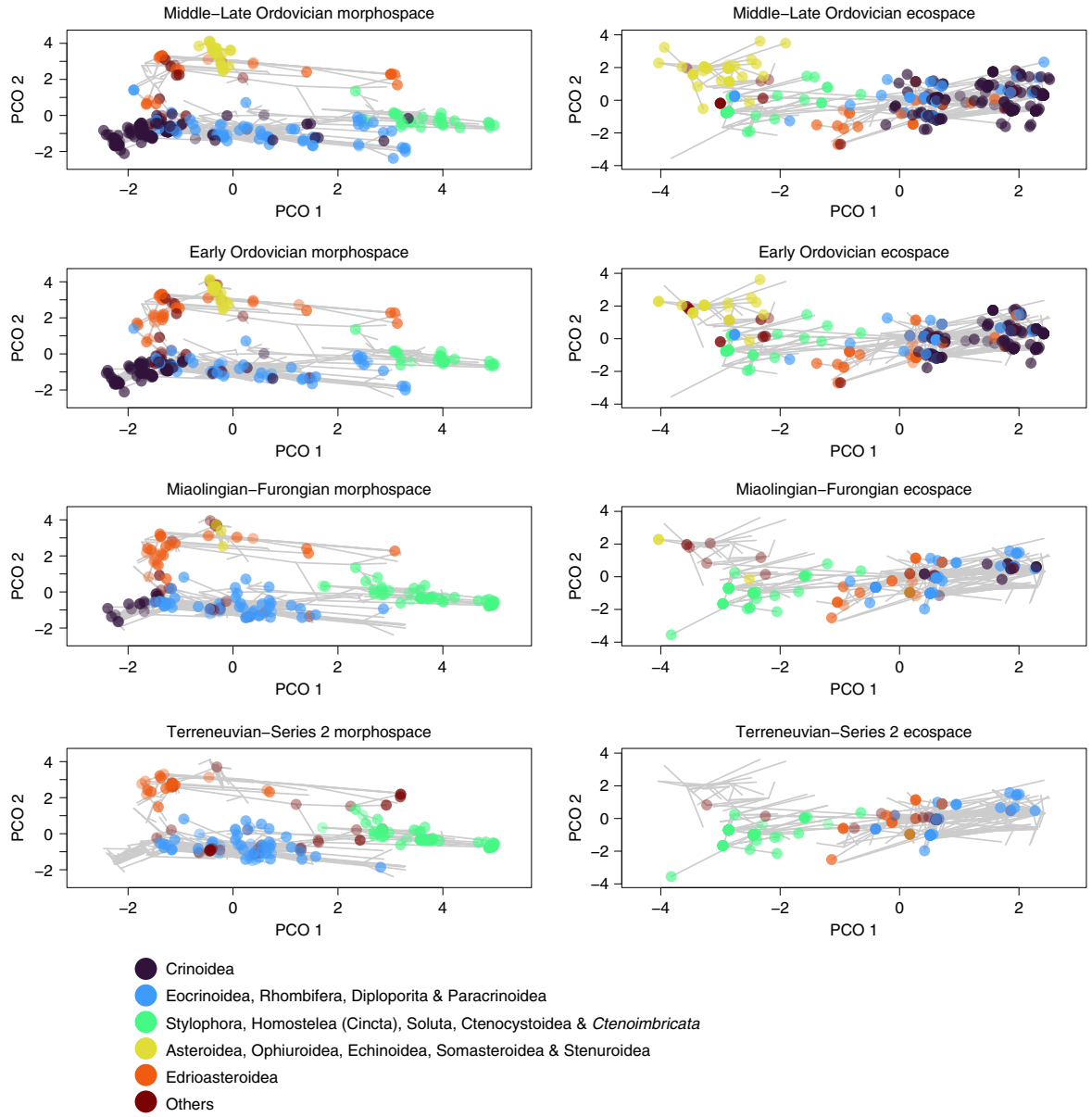



**Fig. 2 | Phylomorphospace and phyloecospace through the Cambrian and Ordovician periods.** First two principal coordinate (PCO) axes are plotted, with each character space combining adjacent epochs, and with points coloured according to taxonomic class. Ancestral-state reconstructions were inferred before ordination to avoid artefacts caused by post-ordination distortion[37]. Points are transparent to illustrate regions where multiple tips and ancestors overlap, with ancestral classes assigned when all descendants are assigned the same class, with nodes illustrated as more transparent than tips. Only the spaces for the 50th time-scaled phylogeny are illustrated as an example, using the 'mode' data treatment for the ecospace; the remaining time-scaled phylogenies produce similar ordination structures. See Extended Data Fig. 4 for interpretation of the ecospace structure. 'Others' include blastoid *Macurdablastus*, coronoids, cyclocystoids, helicoplacoids, ophiocistioid *Volchovia*, parablastoids and taxa of uncertain class affiliation.

potential biases (the difference remains significant in both cases: sample-standardized morphological characters evolve 74–81% faster than ecological ones; Fig. 3a and Extended Data Fig. 6). The morphological dataset contains a greater number of multistate characters, potentially facilitating more opportunities for character change[27]. However, these differences are also negligible between the two datasets (Extended Data Fig. 7). Datasets with greater proportions of missing states (for example, unpreserved, inapplicable or uncertain) can mask character change and reduce rates[30], which would instead magnify the differences observed here: the ecological dataset is nearly complete (91.6–98.6%, depending on the algorithm used to propagate missing states), whereas the morphological dataset is much more incomplete, with 9.1% taphonomically missing states and 62.3% inapplicable states (Extended Data Fig. 8), and in general, there is a greater proportion of missing data in both datasets during the Cambrian, when observed rates are most elevated.

There is substantial variation in the rates at which individual characters evolve (Extended Data Figs. 5 and 7), and the differences are biologically sensible. Significantly fast-evolving anatomical characters—generally involving size and shape of anatomical structures, sculptural ornamentation and number of plates—offer little importance for anatomical function, while the slowest characters—for example, aboral collar, calcareous ring and homoiostelean stele—generally involve taxon-specific synapomorphies defining major clades (Supplementary Table 3). Body size, its life-habit correlates, and mobility are the fastest-evolving ecological characters, whereas

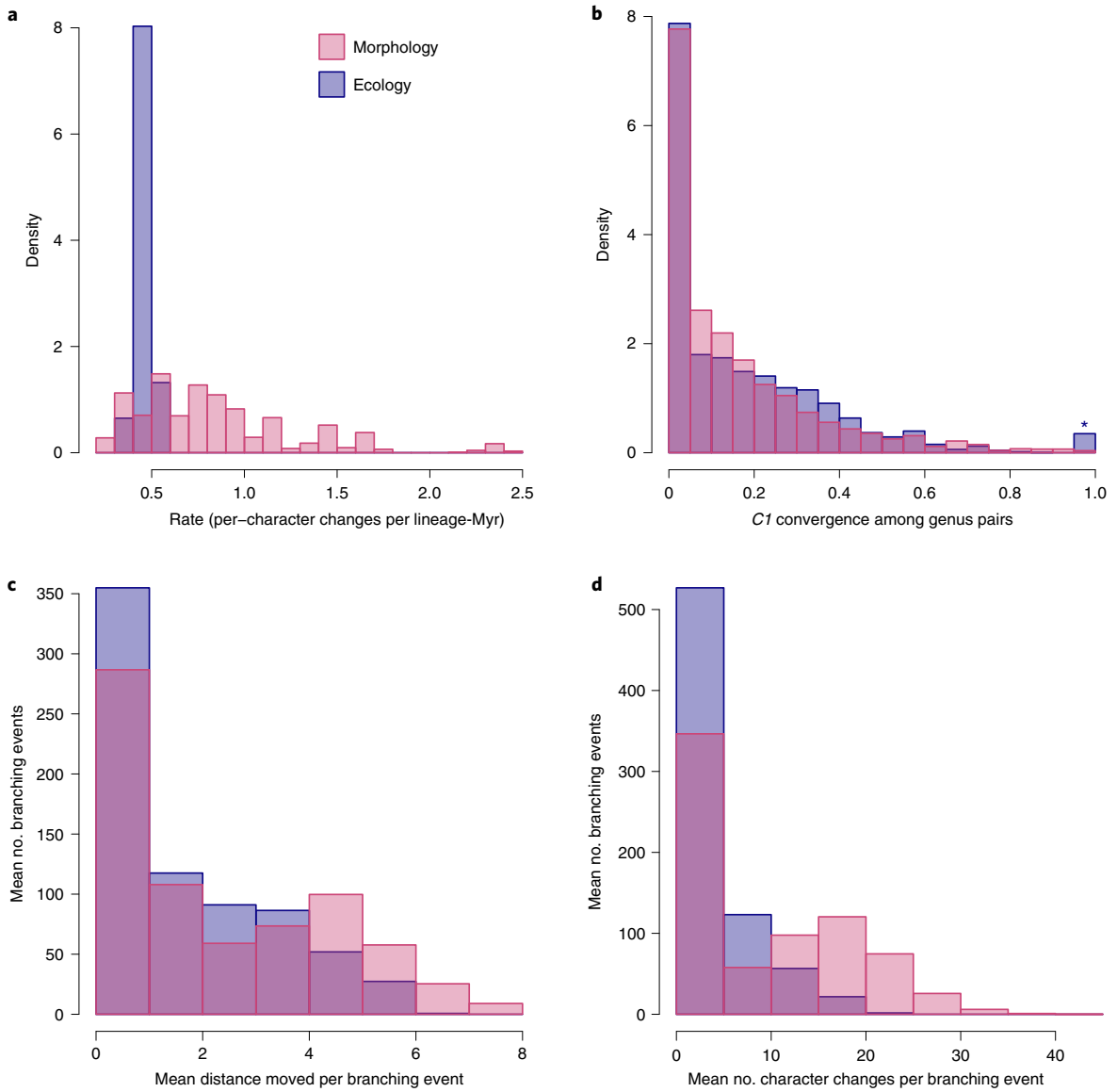



**Fig. 3 | Phylogenetic distributions of divergence and convergence. a**, Subsampled rates of character change (2,000 replicates, drawing 30 characters at random from all 50 time-scaled phylogenies 40 times); after resampling, $\lambda_{\text{morphology}} = 0.837$ character changes per lineage-Myr; $\lambda_{\text{ecology-mode}} = 0.463$; 90.6% of replicates support two-rate model (Akaike weight ≥ 0.90). **b**, Convergence, measured as the most frequent *C1* value for each genus pair across all time-scaled phylogenies. Asterisk denotes the high frequency of convergently identical life habits. Distributions are different (Kolmogorov–Smirnov two-sample test: $D = 0.082$, $P < 2.2 \times 10^{-16}$). **c**, Mean distance moved in character space during phylogenetic branching events (Mann–Whitney *U*-test: $U = 7.82 \times 10^{8}$, $P < 2.2 \times 10^{-16}$). **d**, Mean number of character changes during branching events ($U = 9.02 \times 10^{8}$, $P < 2.2 \times 10^{-16}$). For **c** and **d**, all ancestor–descendant pairs across all 50 time-scaled phylogenies are illustrated, with the ordinate divided by 50 to represent a mean density distribution per phylogeny.

the slowest involve diet, foraging habit and food location, and secondarily reproductive mode and habitat (Supplementary Table 4). These patterns reinforce that body size is more labile than other traits[31], even during the earliest origins of a clade, and that feeding specializations are generally fixed during these origins[7,11]. Although only 21 morphological characters evolve at rates faster than the fastest-evolving life-habit character, the average fast-evolving morphological character evolves 52% faster than fast-evolving ecological characters (17.27 vs 11.38 per-character changes per lineage Myr) (Extended Data Figs. 5 and 7).

**Morphological and ecological disparity are decoupled.** The unfailingly faster rate of morphological change results in distinct disparity dynamics that reflect the anatomical and functional divergence of these earliest echinoderms. Morphological disparity increases earlier than ecological disparity across multiple metrics (Fig. 1c,d and Extended Data Fig. 3), and remains greater throughout the Cambrian. However, relative ecological disparity begins to outpace morphological disparity by the end of the Cambrian, and remains proportionally equal to or greater throughout the Ordovician. These trends are uncorrelated (mean detrended correlation coefficient $R = 0.24$ across nine disparity metrics and three treatments of ecological dataset; Supplementary Table 6), confirming that anatomy and life habits can record distinct histories during evolutionary diversifications.

Despite playing out at a slower tempo, ecological diversification persists unabated across the early Palaeozoic, whereas morphological disparity is divided into distinct period-specific phases (Fig. 1c,d

and Extended Data Fig. 3). The earliest echinoderms had already evolved distinct body plans and life habits, yet Cambrian forms are quite generalized compared with later taxa, occurring in relatively central positions in multivariate character space (morphospace/ecospace) (Fig. 2). This is reflected in their consistently low disparity throughout the Cambrian (Fig. 1c,d), even after accounting for low richness. Morphological forms plotting near the centroid and in several other regions of morphospace are largely extirpated by the Furongian (Fig. 2 and Extended Data Fig. 4), cementing the morphological distinctiveness of the major subphyla[16].

The dominant Cambrian ecospace pattern, in contrast, is one of progressive partitioning of persisting life habits instead of evolving fundamentally new ones. Major life-habit strategies are discrete during the Terreneuvian (Fig. 2 and Extended Data Fig. 4), but later Cambrian taxa evolve progressively intermediate habits, albeit with proportionally less specialization among epifaunal filter feeders than occurs in the Ordovician. This slow pace of functional innovation is the primary reason ecological disparity remains lower than morphological disparity during the Cambrian[32].

The greatest taxonomic turnover in echinoderm history occurred during the early Palaeozoic, involving Cambrian extinctions of two classes and Ordovician crown-group originations of at least eight more (Fig. 2 and Extended Data Fig. 1). Associated with dramatic increases in lineage richness (Extended Data Fig. 1) but a relatively modest increase in rates of character evolution (Fig. 1a,b), the impact of this transition is morphologically profound but ecologically trivial (Fig. 1c,d and Extended Data Fig. 3). Rapid changes in morphological disparity (Fig. 1c,d and Extended Data Fig. 3) result from the addition of new crinoid, asterozoan and echinozoan body plans (Fig. 2 and Extended Data Fig. 4) that expand into novel regions of morphospace. In contrast, this turnover is ecologically invisible because of high degrees of functional convergence (Extended Data Fig. 4). Newly originating echinoderm taxa inhabit minor variations of previously successful strategies rather than truly novel ways of living.

Middle and Late Ordovician echinoderms progressively fill in and expand both ecospace and morphospace, resulting in high levels of disparity and continuing levels of anatomical and functional innovation (Fig. 2). The most extreme morphological forms and ecological functions are occupied by some lineage throughout the entire Ordovician, evident in persistently large convex-hull volumes (Fig. 1d). During the first end-Katian pulse of the Late Ordovician mass extinction, there is a small reduction in overall morphospace, but no such ecospace retraction occurs because only functionally intermediate or redundant taxa go extinct (Fig. 2).

**Phylogenetic structure differentially constrains convergence.** These patterns can be explained by genuine differences in how morphology and ecology are constrained phylogenetically within early echinoderms. Tanglegrams[33,34] confirm that anatomy retains 9.3% greater phylogenetic signal than does life-habit (two-sided *F*-test: $F = 1.093$, $P = 2.16 \times 10^{-9}$), although the difference is minor given the large number of genus pairs tested. This implies greater opportunities for convergence among functional characters than for anatomy, and analyses using the *C1* statistic bear this out. *C1*[35] measures the proportion of character space (morphospace or ecospace) that lineages have evolved independently similar anatomies or life habits: 0 means no tipward evolution has involved convergence and 1 means all character space has been closed by convergence. Although convergence is rare among early echinoderms (affecting < 2% of echinoderm lineages), there are many more instances of ecological convergence (Fig. 3b). Some 295 genera are ecologically strongly convergent with other taxa (that is, $C1 \geq 0.98$), whereas only 31 genera are anatomically so, and there is a substantial phylogenetic influence on this evolution. Morphologically highly convergent genus pairs (that is, $C1 \geq 0.90$) are typically restricted to genera in the same clade (that is, in different taxonomic orders of the same class or subclass; Extended Data Fig. 9), and these pairs are 93.9% convergent, on average. In contrast, when ecological convergence occurs, it not only involves greater degrees of absolute convergence (that is, unrelated genera have identical life habits; mean $C1 = 1.0$), but also involves significantly greater taxonomic distances (that is, they are typically in different classes or subphyla). Put another way, 68.5% of cases of ecological convergence cross taxonomic classes, whereas only 2.3% of morphological pairings do.

Although echinoderm taxonomic classification is increasingly phylogenetically informed, an alternative test focuses on the actual phylogenetic distance that two convergent lineages travelled since their most-recent common ancestor, regardless of their taxonomic affiliation. Here, too, the distributions of phylogenetic distances are distinct (Extended Data Fig. 10); morphologically convergent genus pairs, on average, move twice as far in morphospace during convergent evolution (median = 12.0 distance units) than do convergent pairs in ecospace (median = 5.9 units). This is unlikely due to differences in the numbers of anatomical and life-habit traits, as the phylogenetic distances between all taxa span the same order of magnitude in both character spaces.

**Morphological evolution is more volatile than ecology.** The pattern of convergent evolution in early echinoderms is thus reversed when observing the absolute phylogenetic distances evolved by these genera instead of their taxonomic similarity. Morphological convergence requires a greater phylogenetic distance to occur than does ecological convergence. This apparent paradox is resolved when considering that ecologically convergent pairs had common ancestors with relatively similar life habits, requiring smaller ecological changes to achieve their convergent life habits, whereas the most-recent common ancestor for morphological convergence was more anatomically dissimilar. Most instances of ecological convergence involve taxa with filter-feeding habits, a versatile strategy displayed by a wide range of divergent echinoderm taxa, including crinoids, edrioasteroids, eocrinoids, rhombiferans, diploporitans, paracrinoids and helicoplacoids, sometimes replacing one another through geological time and other times co-occurring as finely partitioned, convergent variants (Fig. 2 and Extended Data Fig. 4). In contrast, most instances of anatomical convergence occur among crinoids. In other words, ecological convergence in these earliest echinoderms involves evolutionary histories dedicated to tinkering within relatively small clusters of ecospace rather than expanding outside them, whereas anatomical convergence is facilitated by substantial changes within a fundamental body plan architecture that ultimately yields a more novel range of morphological combinations[16,36].

This hypothesis was tested by measuring the distance associated with branching events across the phylogeny[37]. If true, the distance moved in ecospace ought to be less than when morphological changes occur. For all branching events in the phylogeny, we tabulated the morphological and ecological distances that each descendant moved away from its ancestor, and the number of character changes that occurred during each event. As predicted, morphological evolution involves a 41% greater evolutionary distance ($\text{median}_{\text{morphology}} = 1.42$ units) during an average branching event than does ecological distance ($\text{median}_{\text{ecology}} = 1.00$; Fig. 3c). Furthermore, there are seven times more morphological character changes than ecological character changes during an average branching event (Fig. 3d; $\text{median}_{\text{morphology}} = 7.0$ character changes, $\text{median}_{\text{ecology}} = 1.0$). Compared with anatomical evolution, functional innovation proceeds at a slower rate and involves smaller changes during branching events.

## Discussion

The earliest echinoderm diversifications demonstrate that anatomy and life habits can provide contrasting but complementary perspectives on evolutionary history[13]. Although ecological opportunity

is widely considered the necessary precursor to anatomical innovation[2,3,36], these results demonstrate that morphological novelty can not only precede ecological success, but that it also involves faster and more frequent anatomical changes, with descendants being more morphologically distinct than their ancestors. These processes are not restricted to the Cambrian radiation, but persist throughout the early Palaeozoic. Although morphological evolution is more volatile, the slow, steady rate of ecological tinkering within a stable adaptive landscape, combined with iterative convergence and modest amounts of expansion within general functional strategies, ultimately allowed early Palaeozoic echinoderms to explore a greater expanse of functional diversity. However, such convergence was restricted to taxa whose ancestors shared similar ways of living, implying greater costs, or fewer opportunities, for major functional changes compared with anatomical novelty[1,2,5].

## Methods

**Compiling and formatting morphological and ecological datasets.** Two fundamental datasets were used in analyses, both collected independently of one another and independent of the informal supertree of 366 Cambrian–Ordovician echinoderm genera. The independence of this information is a critical step in reconciling whether morphology or ecology is the primary driver of a clade's success[38]. Unlike many previous studies that use morphological data compiled originally for different purposes (for example, systematics to identify study taxa, informative characters to produce a phylogeny, the same characters used to interpret both morphological and ecological patterns simultaneously), here we utilized datasets collected by different researchers working independently, choosing characters that represent the breadth of echinoderm anatomy and life habits, and without regard to their potential phylogenetic utility.

The morphological (that is, anatomical) dataset for the same echinoderms was compiled by Brad Deline and colleagues (including co-authors Nicholas Smith, Kathryn Hanson, Rachel Lively, Clayton Collins and Runeshia Parker)[16,39] independently of the cladistic characters used when compiling the supertree. The characters include not only potential autapomorphies, but also symplesiomorphies and other cladistically non-informative characters to describe the comprehensive anatomy of early echinoderms and to avoid ascertainment biases that can affect rate calculations[29,40]. Because it was compiled by a single research group using consistent criteria, it avoids many criticisms of large character datasets[41]. The dataset has 413 discrete characters: 30 numerics, 68 ordered factors, 147 unordered factors and 168 binaries (Supplementary Table 1 and Data 2). Characters span the entire echinoderm body plan, including oral surface, oral opening/frame, oral plates, ambulacra and recumbent ambulacra, floor plates, aboral covering, cover plates, interambulacral plates, erect feeding structures, respiratory structures, periproct, tegmen, theca, pelma, non-feeding appendages, mimicking aboral coverings, and vermiform aboral coverings. Of the characters, 369 are hierarchically dependent (that is, contingent) on other characters, with levels spanning secondary through quinary dependents. Of possible tip states, 28.6% are coded, with 9.1% missing (unknown) and 62.3% inapplicable (that is, number of brachials for an armless echinoderm). When including reconstructed ancestral nodes, there are on average 731 unique morphotypes in this dataset across time-scaled trees, meaning all operational taxa are morphologically distinct.

The ecological (that is, functional or life-habit) dataset has 40 discrete characters (5 numeric, 1 ordered factor, 1 unordered factor and 33 binary characters) representing body size, mobility, reproduction, feeding, resting and feeding habitat, and substrate characteristics[21] (Supplementary Table 2 and Data 3–5). The ecological dataset was compiled by Phil Novack-Gottshall and colleagues (including co-authors Ali Sultan, Jack Purcell and Isa Ranjha). The characters are modified from the framework of ref. [21] to add the ordered-factor character 'filter density' to describe the density of the filter-feeding structure (when present), with states low density (as in edrioasteroids, rhombiferans and most eocrinoids), intermediate density (as in most 'homocrinid' crinoids), and high density (as in most iocrinid crinoids). The use of 40 characters allows a much more sophisticated and functionally nuanced description of echinoderm life habits than previous functional analyses that typically use 3–5 characters to represent the life habits of marine animals[42–44].

Life-habit codings were drawn from 167 references (Supplementary Appendix 1), with body sizes measured from 87 references (Supplementary Appendix 2). Because the ecology of many early echinoderm genera has not been formally studied for all characters used here, missing ecological character states were inferred using three data treatments spanning a range of conservative to liberal assumptions. Because the use of treatment has negligible effect on subsequent analyses and disparity trends are correlated across treatments (Supplementary Table 5), most analyses use the 'mode' treatment that includes the greatest proportion of non-missing data.

The 'mode' treatment is most liberal, and infers missing life-habit states using the most frequent state of (taxonomically) closely related taxa. Of possible tip-taxon states, 98.6% were coded using this treatment, with 1.2% states missing (unknown or unable to be inferred by the 'mode' algorithm) and 0.3% inapplicable (for example, filter density for non-filter feeders). When including reconstructed ancestral nodes, there are 206 unique life habits in this dataset, on average.

The 'constant' treatment is intermediate in degree of assumption between the other two treatments, and infers missing life-habit states only when all (taxonomically) closely related taxa share the identical state. Of possible tip states, 91.6% were coded in this data treatment, with 8.1% missing (unknown or unable to be inferred by the 'constant' algorithm) and 0.2% inapplicable. This data treatment includes 314 unique life habits, on average.

The 'raw' treatment is the most conservative treatment, and only uses life-habit states that are reliably inferred from published literature, making no assumptions about missing data. Of possible tip states, 52.0% were coded in this treatment, with 47.9% missing (unknown) and 0.1% inapplicable. For analyses that trim out taxa with uncoded states, this data treatment includes only 256 genera. In other words, this dataset includes only taxa with coded life-habit characters; it makes no inferences about character states for any tip taxon.

**Analytical workflow: R programming and software.** All analyses were conducted in the R computing environment, v.4.0.2[45], with scripts committed on 29 October 2021 on GitHub at https://github.com/pnovack-gottshall/CamOrdEchinos.

**Building the informal supertree and time-scaled phylogeny.** The phylogeny used for evaluating morphological and ecological trends is an informal supertree of Cambrian–Ordovician echinoderms assembled by Colin Sumrall, Brad Deline and colleagues; see ref. [16] for complete details and image of phylogeny (Supplementary Fig. 1 in ref. [16]). The informal supertree includes 366 echinoderm genera classified in 25 classes, 55 orders and 142 families, representing all known Cambrian genera and most Ordovician genera, plus 5 Silurian/Devonian genera (holothuroids *Palaeocucumaria* and *Porosothyone*, asteroid *Palaeaster*, and ophiocistioids *Rotasaccus* and *Gillocystis*) (Extended Data Fig. 1) to represent members of all echinoderm crown groups.

The overall phylogenetic structure of early echinoderms is still a topic of disagreement, but we utilized an ambulacrarian structure for the tree (for example, see ref. [46]). Most echinoderm taxonomic groups have been treated using modern phylogenetic methods. Several taxa—notably eocrinoids, diploporitans and rhombiferans, and to lesser extent blastoids and asterozoans—are currently considered to be non-monophyletic, although several members of these groups have been placed within monophyletic subclades. The phylogeny used here maintains the phylogenetic structure for well-studied taxa; for taxa that have not yet been included in such analyses, taxa were placed operationally according to their taxonomic affiliation, often within polytomies. However, the only analysis that utilizes taxonomic classification is when identifying the taxonomic distance involved in anatomical and life-habit convergence (that is, Extended Data Fig. 9), which is supplemented with an alternative analysis that uses phylogenetic distance only, regardless of taxonomic decisions (that is, Extended Data Fig. 10). Class-level diversity through time (Extended Data Fig. 1) and character spaces (morphospace and ecospace) for individual taxonomic classes (Fig. 2) are illustrated to provide representative descriptions of temporal trends in morphospace and ecospace structure.

For stratigraphic ranges of tip genera, we downloaded (23 June 2020) ranges from the Paleobiology Database (https://paleobiodb.org/), after uploading relevant references, occurrences, opinions and taxonomic names. Analyses used binning from the 2013 International Commission on Stratigraphy timescale[47] to match chronologic bins used in the Paleobiology Database, although figures displaying a timescale here were modified to incorporate 2020 updates to the interval first appearance datums (FADs) and names (that is, replacing Cambrian Series 3 with Miaolingian).

We used the 'cal3' algorithm[48–51] for building time-scaled trees, using `paleotree::bin_cal3TimePaleoPhy(dataTreatment='firstLast', FAD.only = FALSE)`. The 'cal3' method incorporates not just stratigraphic ranges but also simultaneously models rates of origination, extinction and sampling (preservation). Because the 'cal3' algorithm involves stochastic processes, we built 50 trees to evaluate variability of subsequent results to the tree structure, with 25 trees using `anc.wt = 1` and 25 using `anc.wt = 0` to allow variability in whether taxa can represent tip ancestors. Terminal range additions allowed the time-scaled ranges to be used to estimate phylogenetic diversity. Additional details and justification of rates chosen is provided in Supplementary Information.

Because the 'cal3' method results in fully bifurcating phylogenies, each of the time-scaled trees used here includes the 366 tip taxa noted above, plus 365 ancestral nodes, totalling 731 lineages for the entire dataset used through subsequent analyses. For visualizations of ordination spaces (for example, Fig. 2), we used the 50th time-scaled tree, which was found to most represent a typical 'cal3' tree.

Estimates of time-scaled FADs and last appearance datums are required to analyse temporal trends. We used a modification of `paleotree::phyloDiv()` to use time-scaled branch lengths to obtain node ages (relative to the absolute age of the root time). Ancestral nodes were algorithmically assigned to taxonomic classes when both descendant tips were classified in the same class, and left as 'UNCERTAIN' otherwise.

**Inferring ancestral-state reconstructions.** Ancestral states were inferred separately for the four datasets (that is, morphological dataset and three treatments of the ecological dataset) for each time-scaled tree, using a modification of `Claddis::estimate_ancestral_states()`[26,30] to allow running the reconstructions in a parallel-computing environment. Following best practices[37], we chose the conservative 'pre-OASE 1' implementation that only reconstructs ancestral states when tips are known, and does not subsequently estimate tips with unknown states. (This is especially relevant to the 'raw' data treatment, which retains the large number of uncoded tip characters.) The function uses `phytools::rerootingMethod()`, which wraps partially around the ancestral-character-estimation method `ape::ace()`[52], to calculate the marginal ancestral-state estimate for nodes using Bayesian maximum likelihood[53]. The method uses a continuous-time Markov chain (Mk) model of state changes based on the state-transition matrix *Q*. We used the equal-rate ('ER') model of state changes because we have no previous expectation of variable-rate or symmetric changes. The character matrix was then standardized using the default standardizations in the `Claddis::estimate_ancestral_states()` function concerning handling of uncertain and polymorphic states. Following recommendations[54–56], we included all 366 tip genera and 365 node lineages in subsequent analyses to represent the complete evolutionary history of these echinoderms, with evolutionary durations drawn from the time-scaled trees.

**Converting the character matrices to distance matrices.** We next used `Claddis::calculate_morphological_distances()` to convert the complete character matrices to distance matrices using Wills generalized Euclidean distance (GED)[57], which uses known states in pairwise comparisons to maintain approximately Euclidean distances for missing states. This choice of distance metric introduces a small number of false-positive convergences and idiosyncratic ancestral-state-change histories because of how missing states are estimated in the distance matrix (see discussion in main text and sections below titled 'C1 statistical convergence analyses' and 'Taxonomic distance during highly convergent evolution' for details; also see ref. [58]). However, the relatively few missing states (9.1% for the morphological dataset; 1.2% and 8.1% for the 'mode' and 'constant' treatments of the ecological dataset) minimizes concerns for these analyses[58]. Five time-scaled trees had substantial numbers of uncertain ancestral-state reconstructions for (only) the morphological dataset, which prevented them from producing calculable distance matrices; these five morphological distance matrices were skipped in some subsequent analytics when calculations could not be completed. Because of the hierarchical nature of the datasets, we implemented the 'alpha' method[59] that uses differential weighting of dependent characters (that is, secondary, tertiary and so on, which are only applicable for primary and other independent characters). To balance weighting between dependent and independent characters, we chose alpha = 0.5, in which the primary character contributes weight according to the fraction of shared dependent characters. (Using alpha = 0 would have ignored 89% of the morphological characters and using alpha = 1 would have ignored the 11% primary characters that are consistently coded for all taxa.) Distance matrices calculated using raw Euclidean distance (RED), Gower's coefficient, maximum observable rescaled distance (MORD), and using alternative alpha values were strongly correlated with the Wills GED (alpha = 0.5) matrix, evaluated with a Mantel test using `ade4::mantel.rtest()`[60] with 99 permutations, with minimum pairwise correlation coefficient *r* of 0.645–0.966 across comparisons, and 0.819–0.966 when excluding the least correlated RED. Given such high correlations, it is unlikely that subsequent results would be altered if we had chosen different distance metrics.

**Calculating disparity (and functional diversity) metrics and sample standardization.** We calculated nine disparity metrics for all phylogenetic lineages (nodes and tips) present within stratigraphic stages (ages), of ~5 Myr duration. The metrics included five that are traditionally used when studying morphological disparity and four used when studying functional diversity. We used nine disparity metrics because each is informative in describing different aspects of the dispersion, internal structure (clumping) and evenness with the respective character space (morphospace and ecospace)[4,61,62]. We followed the best-practice recommendations[37,54,63] of emphasizing 'pre-ordination' metrics, which applies for seven of these metrics. Concerns regarding data loss when using post-ordination metrics are lessened here because we used 6 principal coordinates analysis (PCoA) axes—a widely recommended maximum number based on sensitivity analyses[61,64,65]—and only FRic and FDiv use the reduced-dimension data. The metrics used include:

1. *H*: number of unique life habits or morphotypes (trait combinations)[4,66]
2. *D*: mean pairwise Wills GED distance[4,66–68]
3. *M*: maximum pairwise distance[4,66–68]
4. *R*: sum of ranges, the sum of eigenvector ranges in PCoA space[66,69]
5. *V*: total variance, the sum of variances for each character[4,66–68]
6. FRic: functional richness, the minimal convex-hull hypervolume in multidimensional PCoA space across first 'm' PCoA axes[4,70]
7. FEve: functional evenness, the evenness of minimum-spanning tree lengths between taxa in PCoA space[4,70]
8. FDiv: functional divergence, the mean distance of taxa from centroid in PCoA space[4,70]
9. FDis: functional dispersion, the total deviance of taxa from circle with radius equal to mean distance from centroid in PCoA space[4,71,72]

*H*, *D*, *M* and FDis were calculated on the Wills GED (alpha = 0.5) distance matrices; *V* was calculated on the complete ancestral-state matrices; the rest were calculated using PCoA eigenvectors. For the FRic and FDiv metrics that use a subset of PCoA space, we chose m = 6 PCoA axes. We focused on Wills GED (*D*) and convex-hull hypervolume (FRic) for the main text because they represent two widely used disparity metrics and because they describe different aspects of multidimensional dispersion, with *D* describing internal structure and FRic describing external range.

Calculation of disparity metrics was implemented using a modification of `ecospace::calc_metrics()`[63,73], wrapped around a simplified version of `FD::dbFD()`[74], with the following modifications. (1) We directly imported the distance matrix (with ancestral states reconstructed), raw ancestral-state reconstructions and PCoA output to maintain consistency across implementations. (2) We calculated *H* using the distance matrix instead of the character matrix, the former improving handling of missing data. (3) And we rescaled PCoA eigenvectors according to the magnitude of respective eigenvalues;[75] this standardization stabilizes FRic values (and to a lesser extent FEve and FDiv) across subsequent data treatments (that is, sample standardization), so that metrics can be better compared across datasets and subsampling replicates.

Because several disparity metrics are sensitive to sample size[4], we randomly resampled 50 taxa (tips and nodes) during each temporal age to sample-standardize the trends in disparity, and replicated this process 3,000 times per interval, evenly across all time-scaled phylogenies, running analyses in a parallel-computing environment. This number of replicates was chosen on the basis of sensitivity analyses that demonstrate it produces a coefficient of variation (100% × s.d. / mean) within 0.1% across 3–5 test replicates for all intervals and sample sizes. The standard error around each trend was estimated using the standard deviation of the multi-tree resampling distribution[76,77].

**Building phylomorphospace and phyloecospaces using principal coordinates analyses.** We ordinated the distance matrices using PCoA to assess the multivariate distribution of lineages (tips and nodes) in multivariate character space (morphospace and ecospace). We used `ape::pcoa()`, given its wide usage within the palaeontological and functional-ecological literature. In cases where a negative eigenvalue was produced, we implemented the Lingoes correction[78,79]. Because the 'raw' ecological treatment had several taxa with all-missing data, we used `Claddis::trim_matrix()` to remove such taxa from the resulting distance matrices, ancestral-state reconstructions and tree file used for PCoA. The maximum possible number of eigenvalues was 731 because the PCoA was implemented on the 731-taxon distance matrix. After correcting for negative eigenvalues, the median morphological, 'mode' and 'constant' datasets returned 731 usable axes and the 'raw' 513. The mean first PCoA eigenvalue across 50 time-scaled trees represented from 0.66% (for 'raw') to 1.69% (for the morphological dataset) of the relative variance in the datasets. Two disparity metrics (FRic and FDiv) required a subset ('m') of the entire PCoA eigenvector matrix. The optimal recommendation[37,61,64,65] is to use the maximum number of axes that is computationally tractable, and preferably at least 4. We used m = 6 as the best compromise between computational time and a greater number than typically used in functional-diversity analyses[61,65]. Scree plots confirmed that the first six axes represent substantially greater variance than successive axes; the first 6 axes used for calculation of FRic and FDiv account for 1.9% (for constant) to 3.8% (for morphology) cumulative variance. We used `phytools::phylomorphospace()` to superimpose the phylogeny on the resulting morphospace and ecospaces. To assist with interpretation of the resulting character spaces, we used *k*-means clustering of classes using `stats::kmeans()`, and conducted sensitivity analyses using $k = 1$ to 100, choosing $k = 4$ clusters as the optimal balance between interpretability and resolution.

**Calculating rate heterogeneity in character evolution.** We evaluated differences in the rate of character evolution using discrete-character rate tests[27], implementing both the likelihood-ratio test (LRT) and maximum-likelihood-based Akaike information criterion (AIC) method. Both methods incorporate probability models for both branch lengths and character completeness to obtain phylogeny-wide rates of character change ($\lambda$). For the LRT, we used the Benjamini and Hochberg false-discovery rate approach to adjust for multiple comparisons, using corrected alpha = 0.05. For AIC, we compared the two-partition rate model (that is, separate rates for each morphological and ecological dataset) to a single global rate (the same rate shared across both datasets), using Akaike weight to determine whether the two-rate model is better supported statistically than a single-rate model. For both methods, we used a modification of `Claddis::test_rates()` to allow direct input of the character-state matrices (including previously reconstructed ancestral states) calculated above. Rate heterogeneity was evaluated only among the morphological versus ecological character sets, assuming constant rates among clades, individual branches and through time. Given that the phylogeny remains constant for both morphological and ecological datasets, deviations from these

assumptions are unlikely to alter subsequent results. These tests were conducted on each of the 50 time-scaled phylogenies, and resulting statistics are reported as the mean across phylogenies.

Because missing data (whether in the tips or nodes, and whether represented as unknown character states in tip taxa or polymorphisms or uncertain states in ancestral nodes) may influence observed rates of character changes, as well as how confidently disparity metrics reflect evolutionary reality, we also evaluated the nature of missing data in our datasets. We focused on three trends through time in the morphological and ecological sets (using the 'mode' treatment): (1) the proportion of taxa with missing states, (2) the overall proportion of missing states (both across taxa and across characters), and (3) the overall proportion of polymorphisms (that is, 0 and 1) and uncertain states (that is, 0 or 1) in the ancestral-state reconstructions. Because the missing states were not normally distributed across time-scaled trees, we report the trends in median missing data through time.

Because rates can be sensitive to number of characters (and the two datasets have significantly different numbers of characters)[27] and potentially also to temporal variation in number of sampled lineages, we used two subsampling algorithms to evaluate the sensitivity to sampling heterogeneities[77]. The first character-standardizing algorithm evenly sampled a standard number of 30 characters per partition without replacement, across 2,000 replicates, including each of the 50 time-scaled trees 40 times. Characters that lacked variation or occurred in a single species were excluded to focus on observed rates of character change. For the second character-and-taxon-sampling algorithm, we used the same algorithm as the first, but with an additional rule to approximate constant sampling of lineages per stratigraphic age. Because the AIC test calculates $\lambda$ across the entire time-scaled phylogeny and because different lineages have varying stratigraphic ranges, it was not possible to maintain constant numbers of lineages in each bin. Instead, we randomly sampled without replacement, 30 Cambrian lineages and 15 Ordovician lineages per age, which resulted in a mean of 59 lineages per age (median = 66, interquartile range = 51–75) across the resampling distribution of 50 time-scaled trees. Although not precisely constant, this was much more so than the first algorithm, in which lineage richness varied from 5 to 294 lineages per age (interquartile range = 78–204) across the 50 time-scaled trees. To evaluate statistical significance for both subsampling algorithms, we examined the distribution of Akaike weights[80], which provide a measure of the relative support for the one-rate model versus the two-rate model and rate estimates, and we used the conventional cut-off of 0.90 as overwhelmingly strong support for one model against the other.

The LRT method was also used to provide separate rates of evolution for each individual character, and allowed a test of whether the rate of character evolution for individual characters in each dataset was significantly faster or slower than that for other characters. This was repeated for each of the 50 time-scaled trees, and reported as the mean rate for each character. Subsequent statistical analyses, using analysis of variance and two-sample (two-sided) Mann–Whitney test, evaluated differences in whether any identified differences could be influenced by differences in dataset structure in hierarchical character dependents or in numbers of character state. For example, do dependent characters evolve faster than their contingent independent characters? Are there differences in rate among hierarchically different levels? Do datasets with greater number of multistate characters evolve faster?

**Calculating tanglegram offsets.** To explore the phylogenetic structure of the morphospace and ecospace[33], we created tanglegrams (also known as cophylograms) pairing the phylogeny to dendrograms built from the morphological and ecological datasets (here just using the 'mode' ecological treatment). We used `stats::hclust()` to build the dendrograms and `phytools::cophylo()` to optimally rotate the phylogeny to match the dendrograms, repeating for each of the 50 time-scaled trees. We used the two-sided *F*-test to evaluate whether the rank-offsets between each pair of tanglegrams were statistically different across all time-scaled trees, which would indicate differences in degrees of phylogenetic redundancy within each dataset.

**Calculating *C1* convergence statistics.** The *C1* convergence statistic[35] measures the proportion of character space (morphospace or ecospace) that lineages have evolved independently similar anatomies or life habits, with 0 meaning no tipward evolution has involved convergence and 1 meaning all character space has been closed by convergence. To calculate *C1*, we used code in the Supplementary Information of ref. [37], modified for consistency with our previous analyses to use the discrete ancestral-state reconstructions implemented above, and calculating phenotypic distance between tips using the Wills GED (alpha = 0.5) distance matrix calculated earlier (instead of the default raw Euclidean distance). We also calculated the *C3* statistic[35] using similar modifications from ref. [37] but do not report results here because they were very similar to those using *C1*. To evaluate convergence in our datasets (the morphological one and all three treatments of the ecological one), we calculated the *C1* (and *C3*) statistics for all 133,225 pairs of tip genera (= (366 tips − 1)$^2$), for each of the 50 time-scaled trees. Because the distribution of these statistics for each genus pair across time-scaled trees was discrete (for example, 42 trees yielding identical values, with two sets of 4 trees yielding slightly different values), we used the most frequent (modal) statistics across all 50 time-scaled trees for each genus pair in subsequent analyses. We then further examined those taxon pairs with $C1 \geq 0.90$, indicating highly convergent positions in morphospace or ecospace, by calculating their taxonomic distance (measured as the rank that each genus shared the same taxonomic name) and phylogenetic branch distances (measured using a modification of code in `convevol::convrat()`[35] that measures the absolute tipward distance in the Wills GED distance matrix that each genus pair travelled since its most-recent common ancestor).

**Calculating phylogenetic branching (divergence) statistics.** For each branching event in the phylogeny, we tabulated (1) the morphological and ecological distance that each descendant moved away from their ancestor (using the Wills GED distance matrix calculated above), and (2) the number of character changes during each event (using the character-state matrices, including previously reconstructed ancestral states produced above), calculating these statistics for each of the 50 time-scaled trees. In cases of missing states (that is, NA), polymorphisms (that is, 0 and 1), or state uncertainties (that is, 0 or 1), we conservatively only counted changes if the node and tip could not be resolved to the same state. (This is the same logic used when calculating the morphological and ecological distances above.)

**Reporting Summary.** Further information on research design is available in the Nature Research Reporting Summary linked to this article.

## Data availability

Raw datasets are appended as Supplementary Data 1–5. Additional ancillary datasets, R script, time-scaled phylogenies and intermediate R workflow objects have been reposited in the open-source Harvard Dataverse (https://dataverse.harvard.edu/dataverse/CamOrdEchinoderms), under CC0 public domain license.

## Code availability

All code (written in the programming language R) is available publicly at https://dataverse.harvard.edu/dataverse/CamOrdEchinoderms, under CC0 public domain licence.

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## Author contributions

P.M.N.-G. and B.D. designed the study; P.M.N.-G. conducted analyses; P.M.N.-G., B.D. and C.D.S. wrote the paper; P.M.N.-G., B.D., C.D.S., A.S., N.S.S., J.P., K.E.H., R.L., I.R., C.C. and R.P. collected data.

## Competing interests

The authors declare no competing interests.

## Additional information

**Extended data** is available for this paper at https://doi.org/10.1038/s41559-021-01656-0.

**Supplementary information** The online version contains supplementary material available at https://doi.org/10.1038/s41559-021-01656-0.

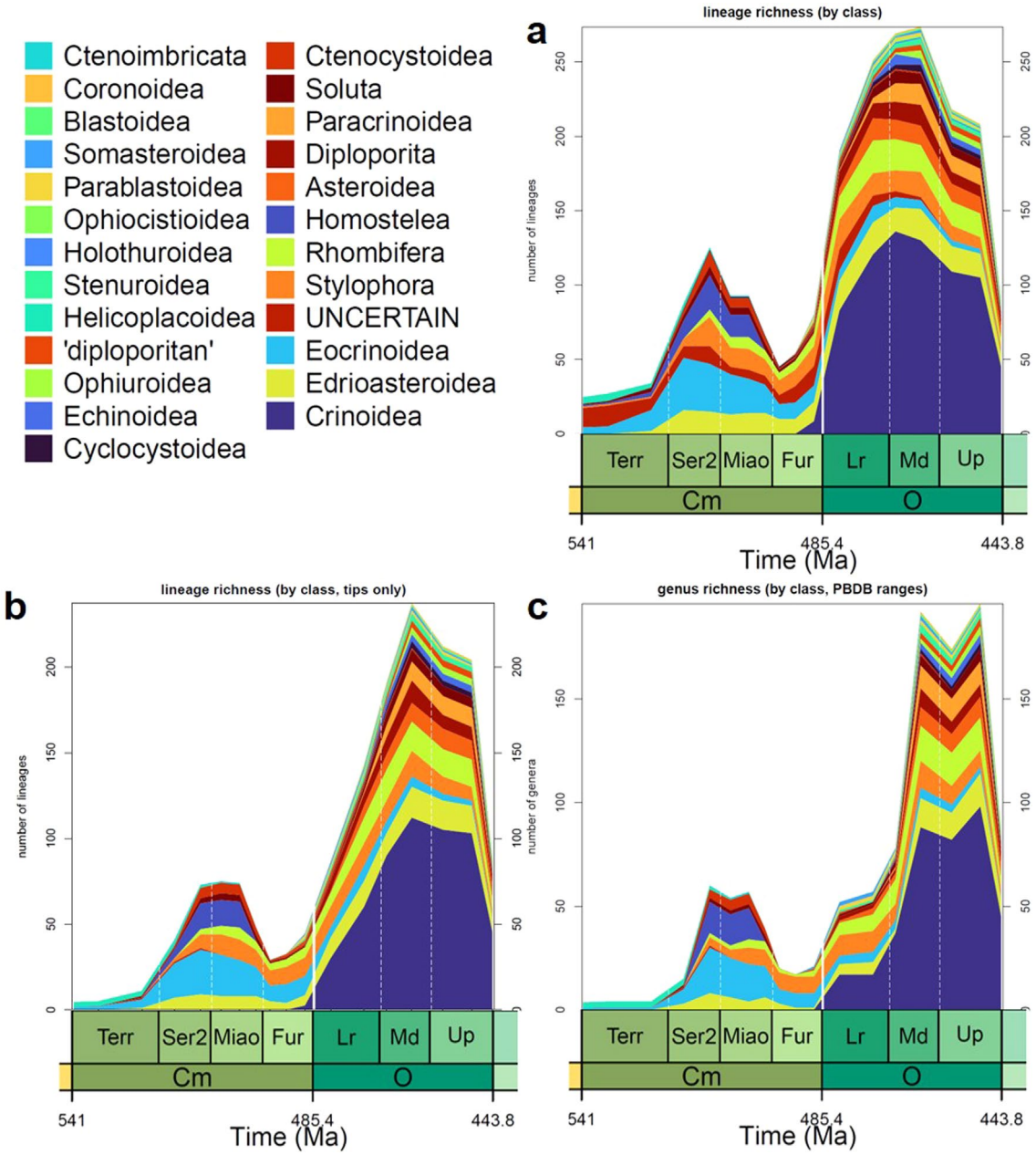
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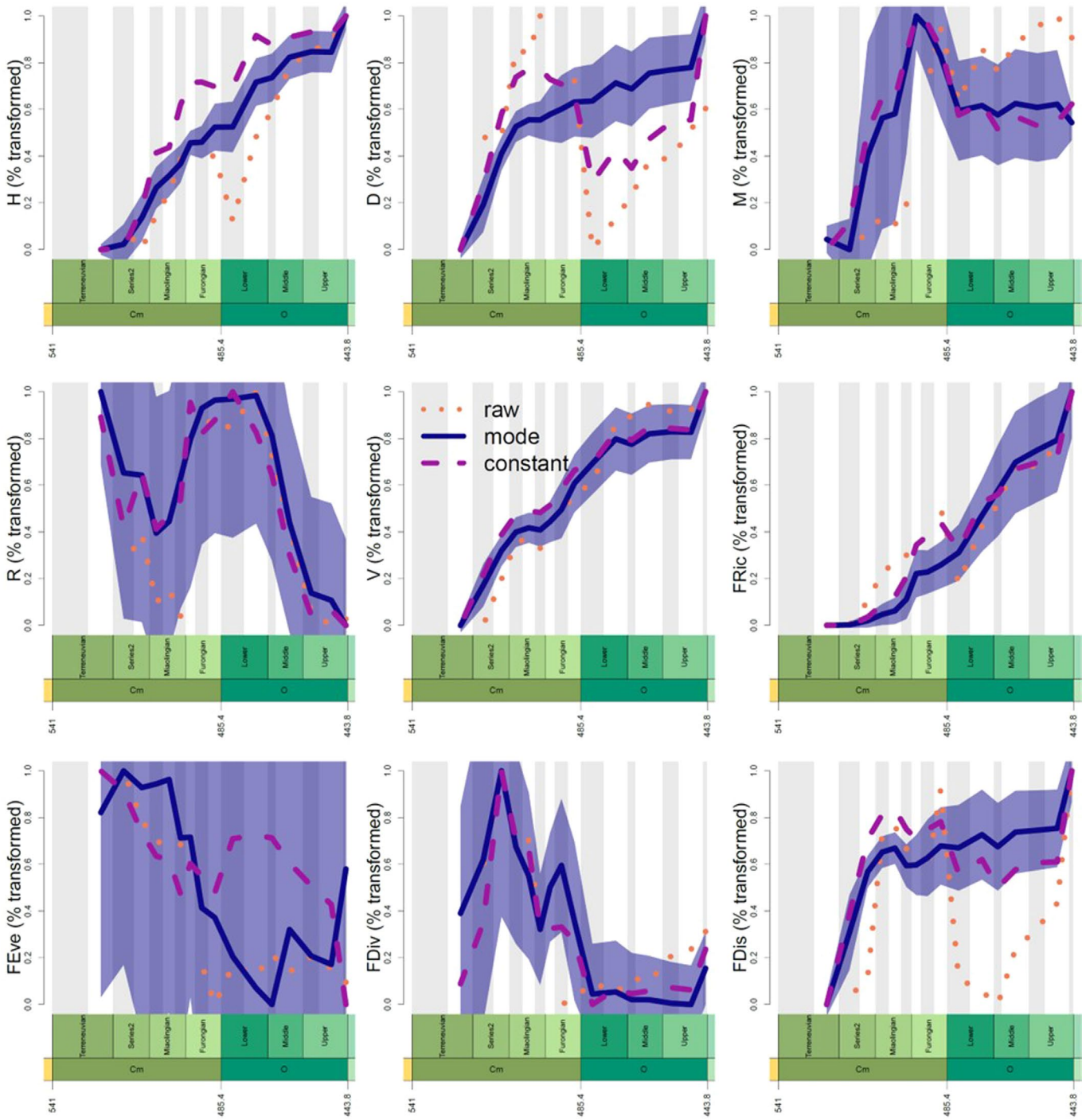
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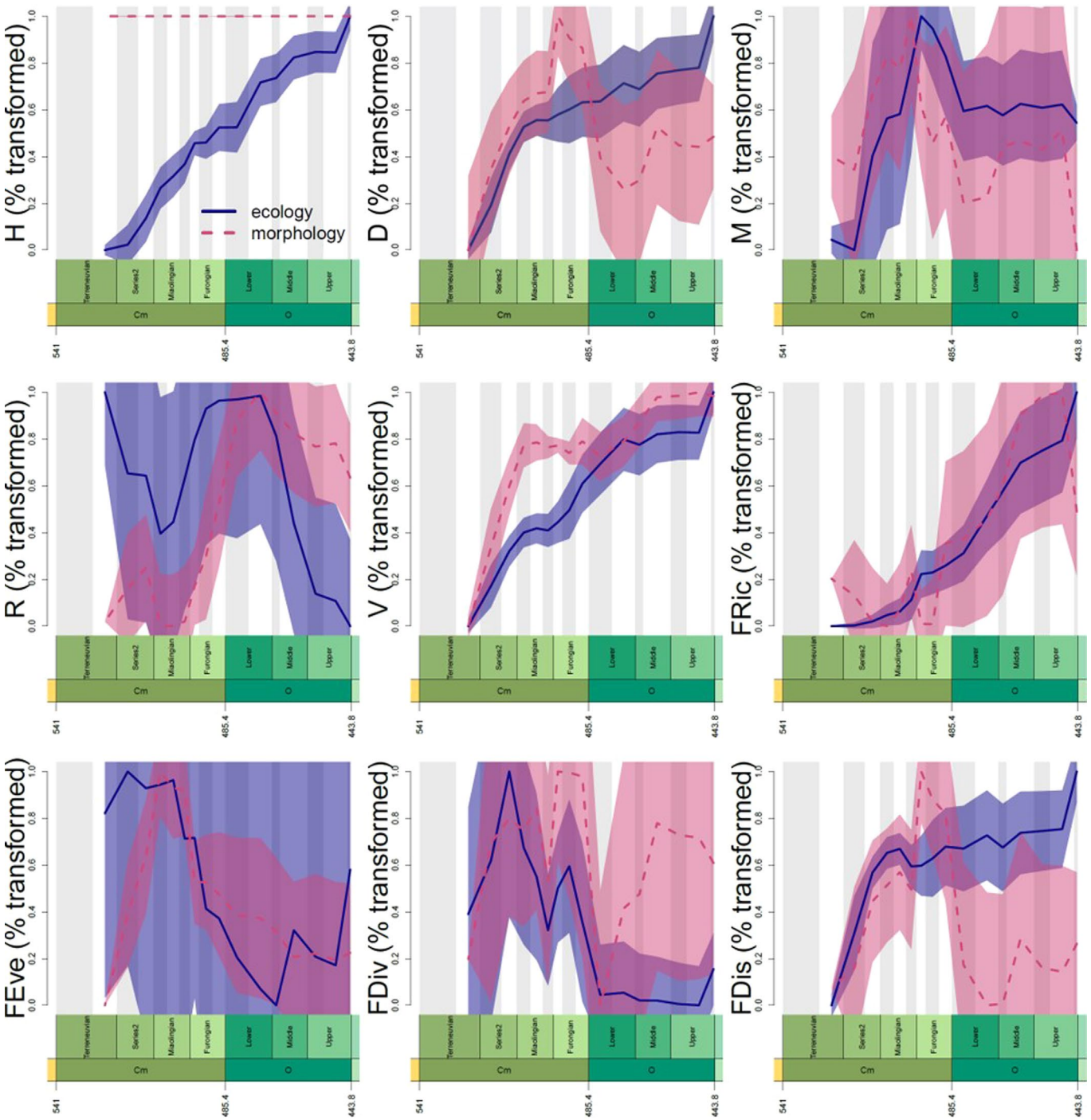
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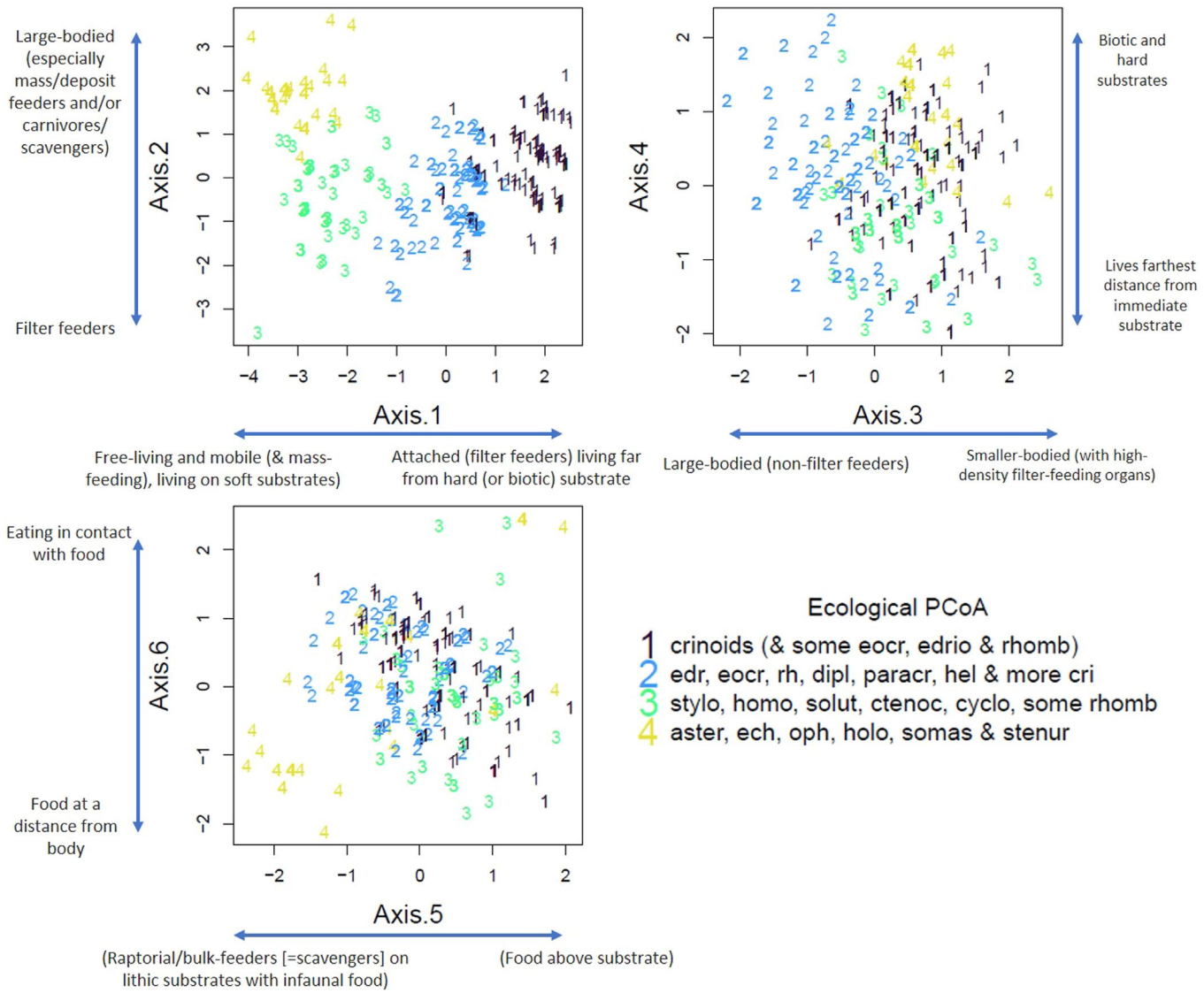

**Extended Data Fig. 1 | Stacked richness trends, color coded according to operational taxonomic class.** a, Tree-based estimate of lineage richness measured using mean of 366 tip and 365 node ranges inferred across time-scaled phylogenies. b, Tree-based estimate of lineage richness measured using mean of only 366 tip-taxon ranges inferred from time-scaled phylogenies. c, Richness measured using only raw stratigraphic ranges from Paleobiology Database (PBDB). Ancestral nodes were assigned to taxonomic classes when both descendent tips were classified in the same class, and left as "UNCERTAIN" otherwise. For comparative purposes, these figures provide alternative representations of lineage richness through time. The legend is ordered to correspond to same order as in the stacked curves, with most-diverse (crinoids, primarily Ordovician) at lower right, and least diverse (*Ctenoimbricata*, early Cambrian) at upper left.



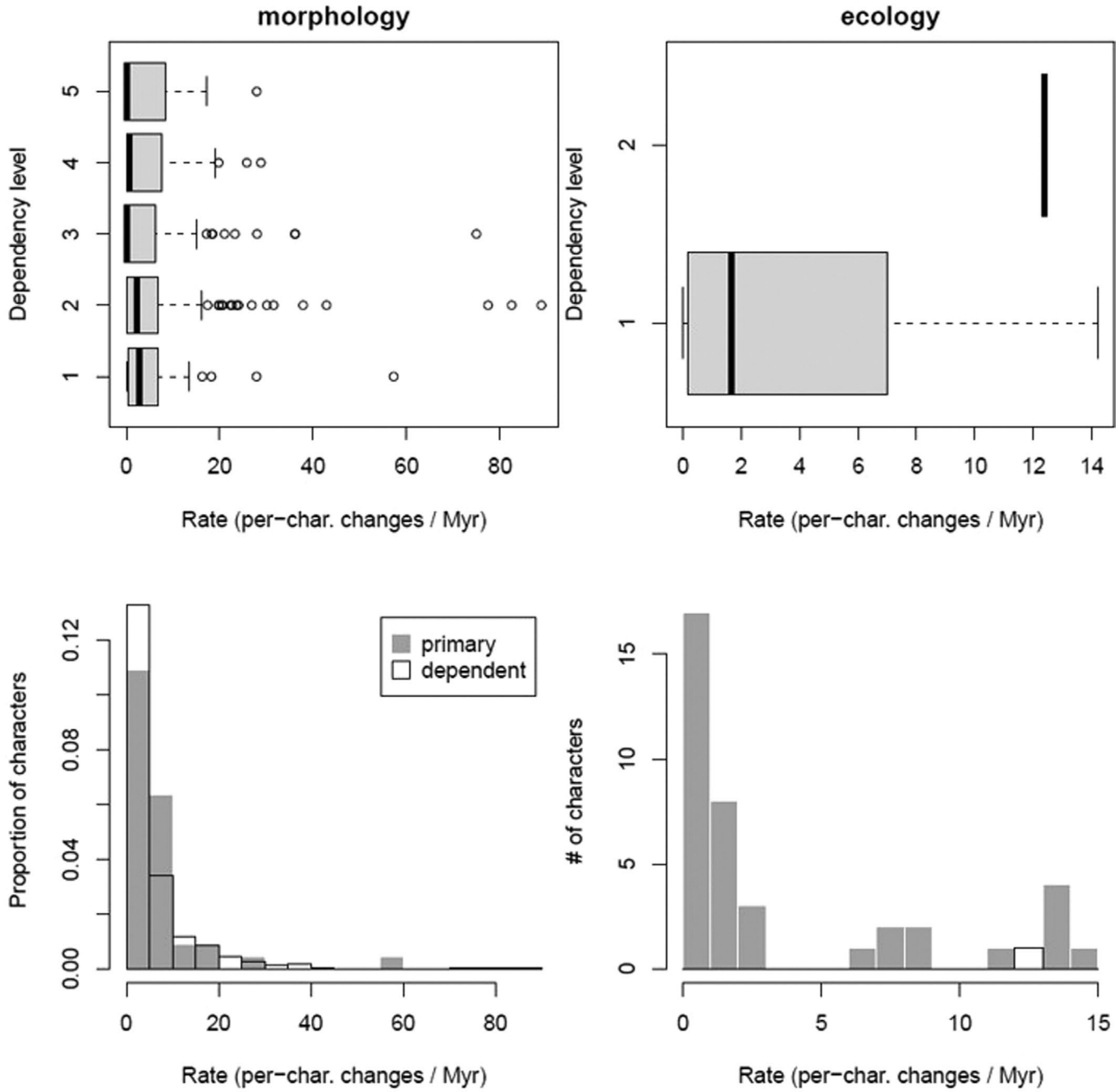

**Extended Data Fig. 2 | Sample-standardized mean trend in nine disparity metrics for the three life-habit propagation treatments (mode, constant, and raw) of the ecological data set.** Sample standardization used a sampling quota of 50 lineages and 3,000 replicates per stratigraphic age (sampling 50 time-scaled phylogenies 60 times each). Trends are percent-range-transformed to focus on shape of trends rather than absolute values. The standard deviation across the multi-tree resampling distribution for the 'mode' treatment is illustrated as a confidence polygon; the variability for the other treatments are of a similar magnitude. Metrics include number of unique life habits / morphotypes (trait-combinations) (H), mean pairwise Wills GED distance (D), maximum pairwise distance (M), sum of ranges (R), total variance (V), functional richness, or minimal convex-hull hypervolume (FRic), functional evenness (FEve), functional divergence (FDiv), and functional dispersion (FDis). See text for description of each metric. See Supplementary Table 5 for statistical correlation tests.



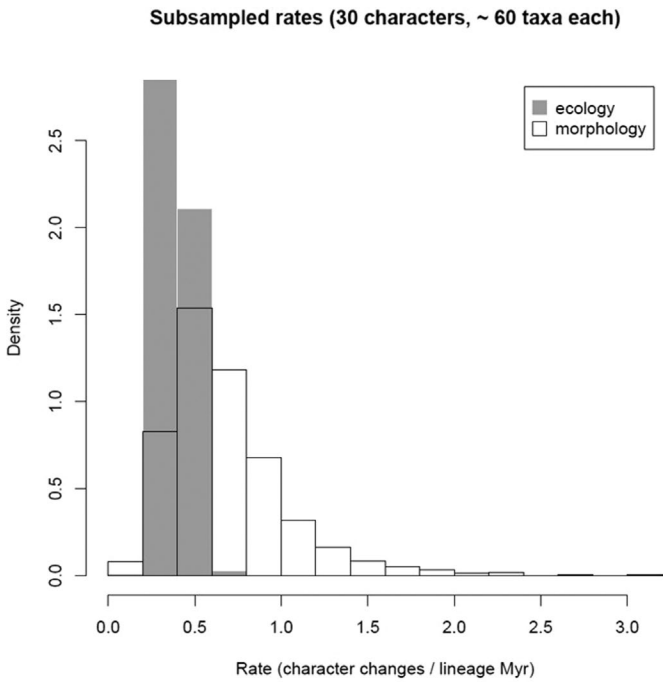

**Extended Data Fig. 3 | Sample-standardized mean trend in nine disparity metrics for the morphological and ecological ('mode' treatment) data sets.** Sample standardization used a sampling quota of 50 lineages and 3,000 replicates per stratigraphic age (sampling 50 time-scaled phylogenies 60 times each). Trends are percent-range-transformed to focus on shape of trends rather than absolute values. The standard deviation across the multi-tree resampling distribution for each trend is illustrated as a confidence polygon. See Extended Data Fig. 2 for abbreviations of each metric. Trends are not illustrated for the 'constant' life-habit treatment. See Supplementary Table 6 for statistical correlation tests (including for the 'constant' treatment).



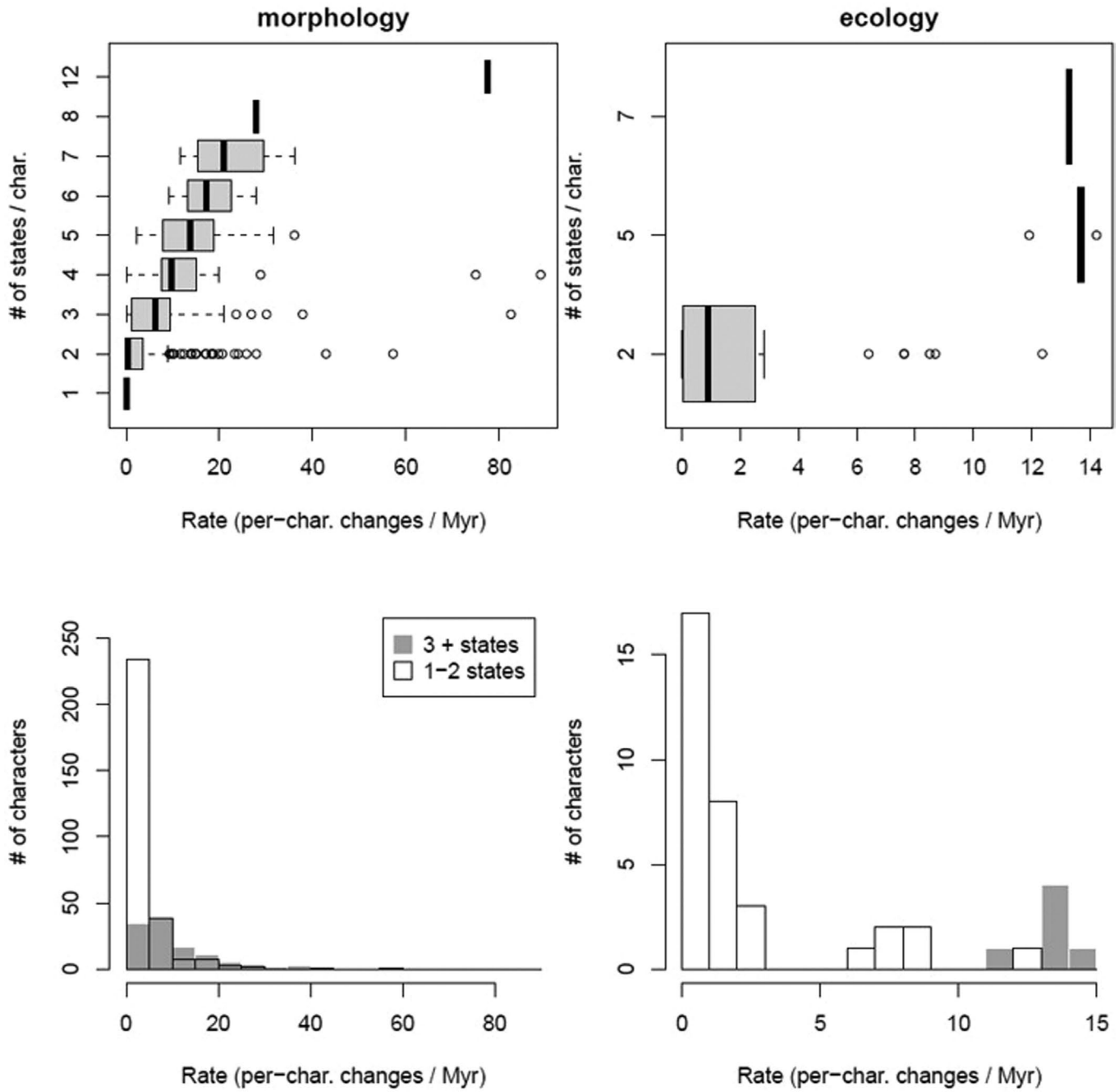

**Extended Data Fig. 4 | Ecospace for first six PCoA axes (using the 'mode' data set), superimposing the k-means clusters and listing the taxonomic classes that are most frequent in each cluster.** PCoA output only illustrated for fiftieth time-scaled tree as one example; remaining time-scaled phylogenies produce similar ordination structures. Life-habit variables that are strongly correlated (positively or negatively) with each PCoA axis are illustrated. Cri = Crinoidea, Eocr = Eocrinoidea, Edrio and Edr = Edrioasteroidea, Rhomb = Rhombifera, Diplo = Diploporita, Paracr = Paracrinoidea, Hel = Helicoplacoidea, Stylo = Stylophora, Homo = Homostelea (order Cincta), Solut = Soluta, Ctenoc = Ctenocystoidea, Cyclo = Cyclocystoidea, Aster = Asteroidea, Ech = Echinoidea, Oph = Ophiuroidea, Holo = Holothuroidea, Somas = Somasteroidea, and Stenur = Stenuroidea. Additional classes with few genera not indicated. In general, these clusters correspond to four major life habits: Group 1 includes sedentary, epifaunal, high-density filter feeders on hard or biotic substrates; group 2 includes motile filter-feeders in soft-substrates, group 3 includes free-living and mobile epifaunal filter feeders on soft substrates; and group 4 includes deposit-feeders, carnivores, and scavengers. Although colors are chosen to match the classes illustrated in Fig. 2, note the points here correspond to the k-means life-habit clusters instead of taxonomic classes.



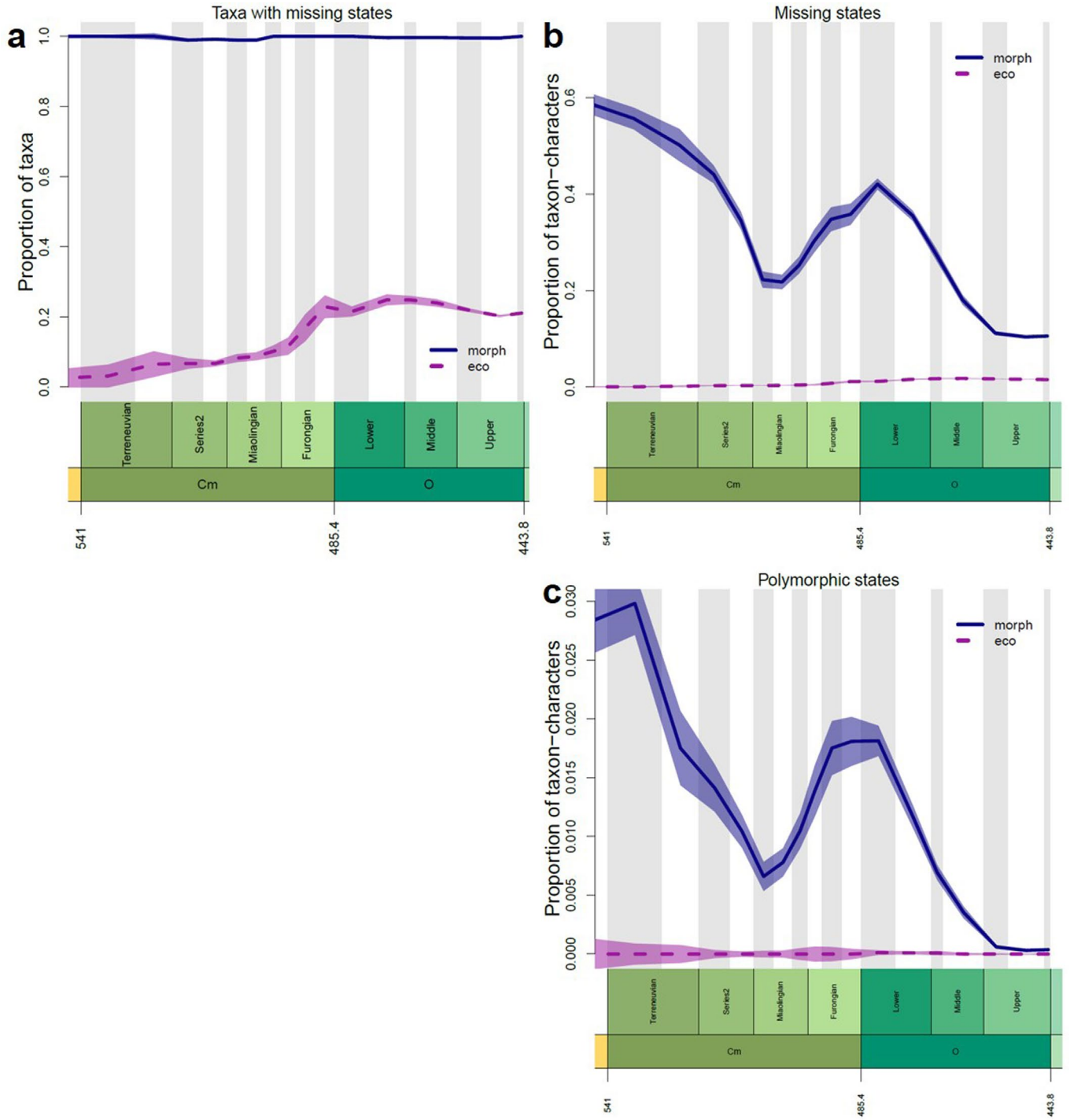

**Extended Data Fig. 5 | Rate distributions for morphological and ecological data sets, according to hierarchical character-level.** a, Per-character rate barplots (box-and-whisker plots) for morphological characters, separated according to hierarchical level. Center line is median, box limits are upper and lower quartiles; whiskers are 1.5x interquartile range; points are outliers. b, Same as panel a, but for ecological characters, which only has a single secondary character (filter density). c, Histogram of per-character rates for morphological characters, separated according to primary versus dependent (secondary through quinary) characters. d, Same as panel c, but for ecological characters. Note that panels c and d have different ordinate scaling to better illustrate the range of character rates, and panel c is reported as the proportion of characters instead of raw numbers. Per-character rates are the mean rate for each character across 50 time-scaled phylogenies.



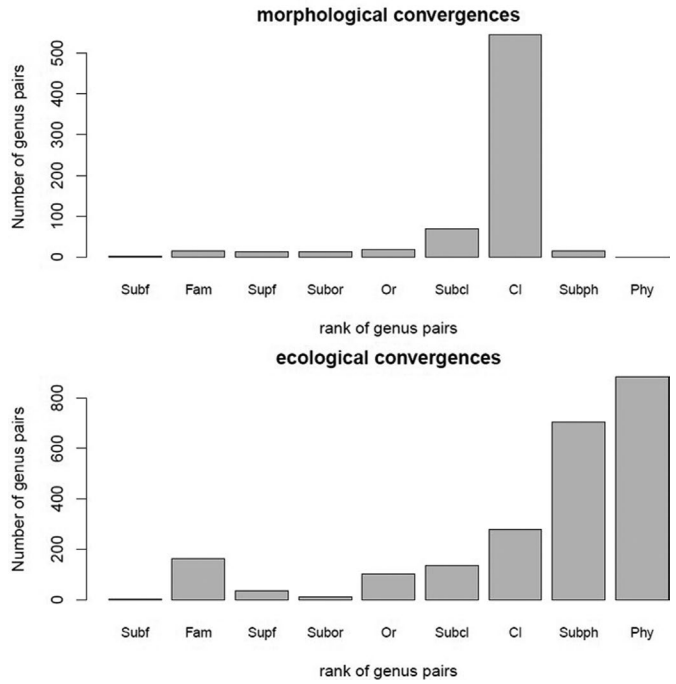

**Extended Data Fig. 6 | Subsampled rate distribution for morphological and ecological data sets, after simultaneously standardizing for number of characters and lineages per stratigraphic age.** The subsampling algorithm drew, at random and without replacement, 30 characters and an average of 59 lineages per stratigraphic bin, evenly sampling all 50 time-scaled trees. After resampling, $\lambda_{morphology} = 0.679$ and $\lambda_{ecology} = 0.390$. 59.1% of replicates support (Akaike weight $\geq 0.90$) that the morphological and ecological data sets evolve at different rates, with the median support equal to 0.986.



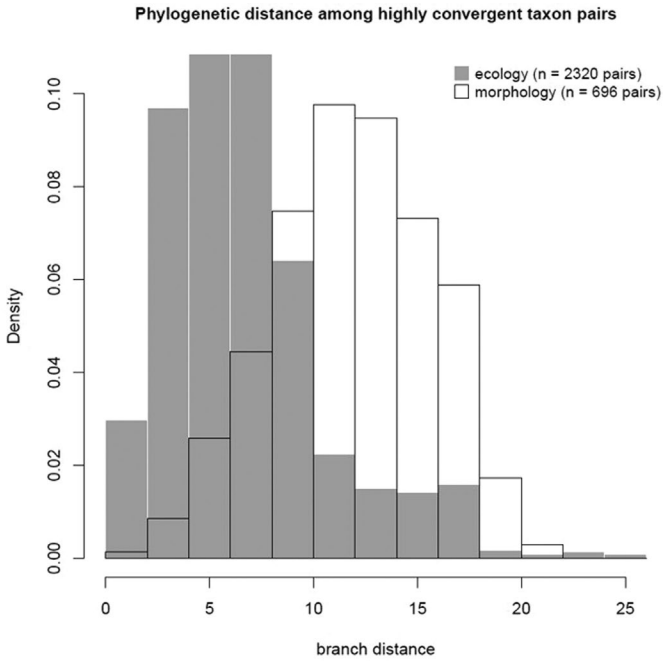

**Extended Data Fig. 7 | Rate distributions for morphological and ecological data sets, according to numbers of character states.** a, Per-character rate barplots (box-and-whisker plots) for morphological characters, separated according to number of states for characters. Center line is median, box limits are upper and lower quartiles; whiskers are 1.5x interquartile range; points are outliers. b, Same as panel a, but for ecological characters. c, Histogram of per-character rates for morphological characters, separated according to characters with 1–2 states versus those with 3 or more states. d, Same as panel c, but for ecological characters. Note that panels c and d have different ordinate scaling to better illustrate the range of character rates. Per-character rates are the mean rate for each character across 50 time-scaled phylogenies.




**Extended Data Fig. 8 | Trends in missing data through time for the morphological and ecological data sets, per stratigraphic age.** a, Proportion of lineages (tips and nodes) with at least one missing character state through time. b, Proportion of states, across characters and lineages, through time. c, Proportion of polymorphic (for example, 0&1) or uncertain (for example, 0/1) states through time. The trend is the median values in each interval across 50 time-scaled trees and the error polygon is the standard deviation across these time-scaled trees.




**Extended Data Fig. 9 | Barplots of the taxonomic rank at which all highly convergent (*C1* ≥ 0.90) genus pairs are classified.** When a convergent pair shares a taxonomic class, Crinoidea for example, it means each genus is classified in the same class but in a different subordinate rank of that class, such as in different subclasses or orders. a, Taxonomic similarity for the morphological data set. b, Taxonomic similarity for the ecological data set. *C1* for each genus pair was calculated as the modal (most frequent) *C1* statistic for that pairing across all 50 time-scaled phylogenies.




**Extended Data Fig. 10 | Proportional distributions of branching distances among highly convergent genus pairs.** Distances were calculated as the sum tipward distance each genus traveled in Wills GED distance space since their most-recent common ancestor (Kolmogorov-Smirnov two-sample test: $D = 0.578$, $P < 2.2e\text{-}16$). Branching distance for each genus pair was calculated as the most frequent distance for that pairing across all 50 time-scaled phylogenies.

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| ☐ | ☒ | The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☒ | ☐ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
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| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ | Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated |

*Our web collection on statistics for biologists contains articles on many of the points above.*

## Software and code

Policy information about availability of computer code

| | |
|---|---|
| Data collection | Genus stratigraphic ranges downloaded from the Paleobiology Database, https://paleobiodb.org/, downloaded June 23, 2020 |
| Data analysis | All code reposited at https://github.com/pnovack-gottshall/CamOrdEchinos. Code uses R (4.0.2) and packages ade4 (1.7-17), ape (5.5), beepr(1.3), Claddis (0.6.3), devtools (2.3.0), doParallel (1.0.16), FD (1.0-12), geiger (2.0.7), geometry (0.4.5), geoscale (2.0), paleotree (3.3.25), parallel (4.1.0), phangorn (2.7.0), phytools (0.7-83), plotrix (3.8-1), Rcpp (1.0.6), snowfall (1.84-6.1), vegan (2.5-7), viridisLite (0.4.0). |

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- A description of any restrictions on data availability
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Raw data sets are appended for reviewer's access. All code, data sets, and phylogeny has been reposited in the open-source Harvard Dataverse (https://dataverse.harvard.edu/dataverse/CamOrdEchinoderms).

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Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

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# Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

| | |
|---|---|
| Study description | Time-series analysis of fossil taxa in a comparative phylogenetic analysis. Morphological (anatomical) and ecological (life habit) data and phylogenetic supertree based on descriptions in published literature surveys. |
| Research sample | 366 Cambrian and Ordovician echinoderm genera (including 5 Silurian/Devonian genera for establishing polarity). Stratigraphic ranges downloaded from Paleobiology Database (https://paleobiodb.org/, downloaded June 23, 2020). Morphological (anatomical) and ecological (functional) traits for all genera treated as separate variables. |
| Sampling strategy | Study includes all Cambrian genera and the majority of Ordovician genera (representing all known lineages). |
| Data collection | Anatomical data compiled by B.D. with N.S., K.H., R.L., C.C., and R.P. Life habit data compiled by P.N.-G., A.S., J.P., and I.R. Supertree assembled by C.D. and B.D. |
| Timing and spatial scale | Morphological data set compiled 2013-2022. Ecological data sets compiled 2015-2020, based on earlier database compiled periodically since 1999. Phylogeny compiled 2014-2019. Stratigraphic ranges downloaded in 2020. |
| Data exclusions | No data were excluded from analyses. |
| Reproducibility | R code provided allows reproducibility of all analyses, including random-number-generator seeds for analyses using Monte Carlo algorithms. |
| Randomization | Random-number-generator seeds are provided in R code for analyses using resampling, subsampling, randomization, or other algorithms using Monte Carlo processes. |
| Blinding | Blinding was not relevant to this study. |

Did the study involve field work? ☐ Yes ☒ No

# Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

## Materials & experimental systems

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ Antibodies |
| ☒ | ☐ Eukaryotic cell lines |
| ☐ | ☒ Palaeontology and archaeology |
| ☒ | ☐ Animals and other organisms |
| ☒ | ☐ Human research participants |
| ☒ | ☐ Clinical data |
| ☒ | ☐ Dual use research of concern |

## Methods

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ ChIP-seq |
| ☒ | ☐ Flow cytometry |
| ☒ | ☐ MRI-based neuroimaging |

## Palaeontology and Archaeology

| | |
|---|---|
| Specimen provenance | Not relevant to this study. Using publicly available database data, combined with published literature surveys and museum collections. |
| Specimen deposition | Not relevant to this study. |
| Dating methods | No new dates are provided. |

☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.

| | |
|---|---|
| Ethics oversight | No ethical approval was required because there are no human subjects used in study and no field collecting of specimens was involved. Access to museums and institutional libraries were provided by affiliated institutions of the authors. |

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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